

4. A recurrent network has some of its outputs connected to its own inputs, so that the outputs at time t are fed back into the network at time $t + 1$. This can be a different way to deal with time-series data. Modify the MLP code so that it acts as a recurrent network, and test it out on the Palmerston North ozone data on the book website.

In this case, we change the MLP into a simple recurrent network by using the previous output as an input for the next time step.

The new input is:

$$[x(t), x(t-1), \dots, y(t-1)]$$

The feedback connection uses a fixed weight of 1, so this weight is not updated during training.

The backpropagation and weight update formulas can stay the same. However, the output value y_k will be different because it now depends on the previous output. Because of this, the error, delta values, and weight values will also be different.

The data must be processed in time order because the output from one step is used in the next step.

Therefore, the main change is adding the previous output to the input of the network. The training equations do not need a new formula for this simple recurrent model.